Department of Mechanical Engineering
Galgotias University

AERODYNAMICS
BAUT3063

Topic- Boundary Layer on Flat Plate

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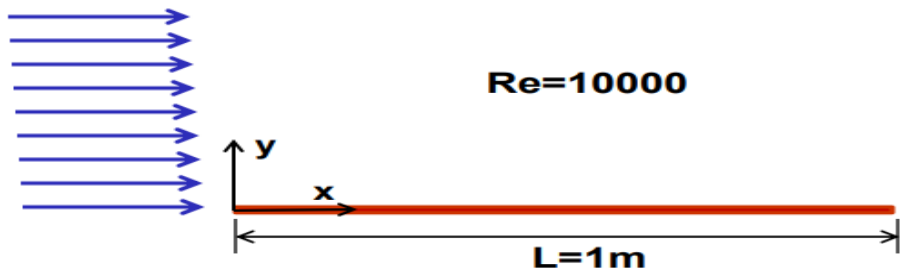
Course: B. TECH AE

I. Introduction

- **Aim:** Solution for boundary layer

- **Problem Specification**
  For the case presented bellow take into account a fluid flow on a flat plate. The Reynolds number based on the plate length is Re=10,000 and the plate length is L=1 m. For simplicity inlet fluid velocity Vo in x-direction is 1 m/s and the fluid density is 1.0 kg/m^3, viscosity m=10^{-4} Pas.
  For presented case solution is only Reynolds number dependent:



- **Methodology**
  1. Creating 2D simple geometry - Spaceclaim
  2. Creating 2D mesh – Mesh
  3. Solver Set-up
  4. Results
  5. Plots
  6. Solution verification

II. CFD Process

- **Geometry:** At this step we will draw simple geometry – we will just plot and set-up dimension for rectangular.
  Process involved: Plane → Sketch → Rectangular → Dimension V1 & H1 → Surface to Sketch.
  Body of 2D Plate is ready.

- **Meshing:** At this step numerical mesh will be created. Mesh is required in order to performed computer simulations – because of methodology used. Continuous space will be replaced by the discrete space.
  Process involved: Physics Preferences = CFD, Solver Preferences = Fluent
  Edge Selection → RMB → Insert Sizing → Details of Edge Sizing → Number of Division = 50 → Behavior = Hard → Bias Type= ___ __ _ _ → Bias Factor = 70
  Face Selection → RMB → Insert Mapped Face Meshing. In Details of "Mapped Face Meshing", Apply to make this surface the geometry selection. This procedure creates uniform structural mesh. **Generate Mesh**

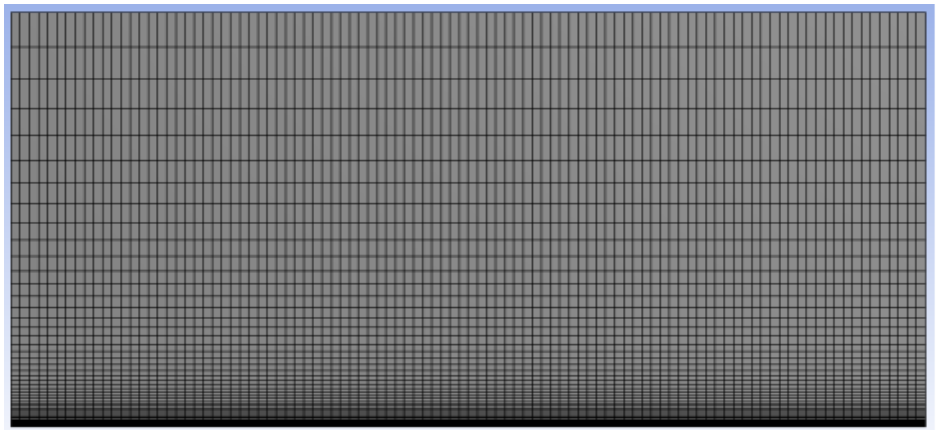
- **Solver-Setup:** Fluent Solver is assigned the following parameters to determine the various results.
  Fluent Solver → 2D → Double Precision → Serial → Ok
  Mesh → Check → Info > Size → Info > Quality → Info > Memory Use → Display
  General → Steady → Planar → Pressure-Based → Ok
  Model → Viscous – Laminar → Ok
  Material → Air (Density=1, Viscosity=0.0001) → Create/Edit → No
  Boundary Conditions:
  Inlet → Velocity Inlet → Velocity Magnitude = 1 → Ok
  Outlet → Pressure Outlet → Gauge Pressure = 0 → Ok
  Plate → Type= Wall → Wall Motion = Stationary Wall → Shear Condition = No Slip → Ok
  Cell Zone Condition → Fluid = Air1 → Ok
  Solution Initialization → Standard Initialization → Compute = Inlet → Gauge Pressure 0 Pa, X-Velocity = 1 m/s and Y-Velocity = 0 m/s → Initialize
  Run Calculation → Iterations = 1000 → Calculate

III. Result Analysis and Discussion

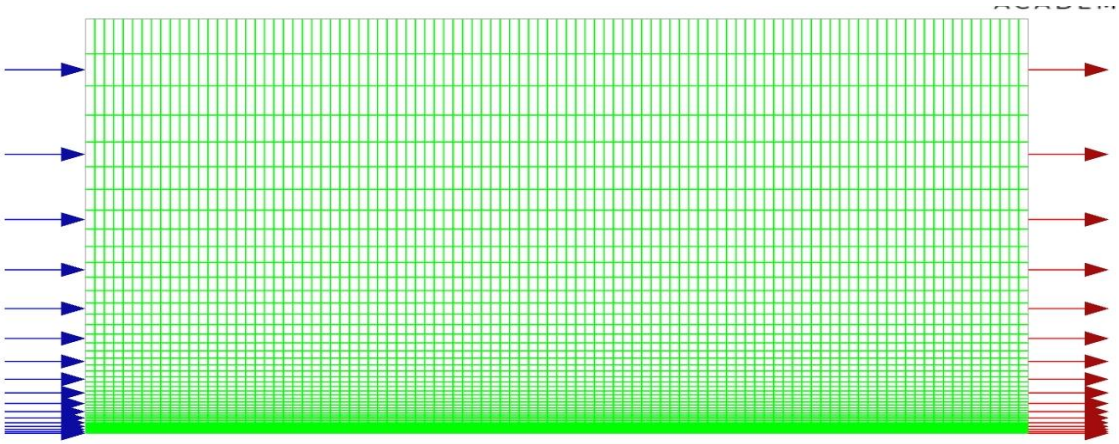
❖ Model

Rectangular Plate of Length 0.5mm and Breadth 1mm

❖ Mesh

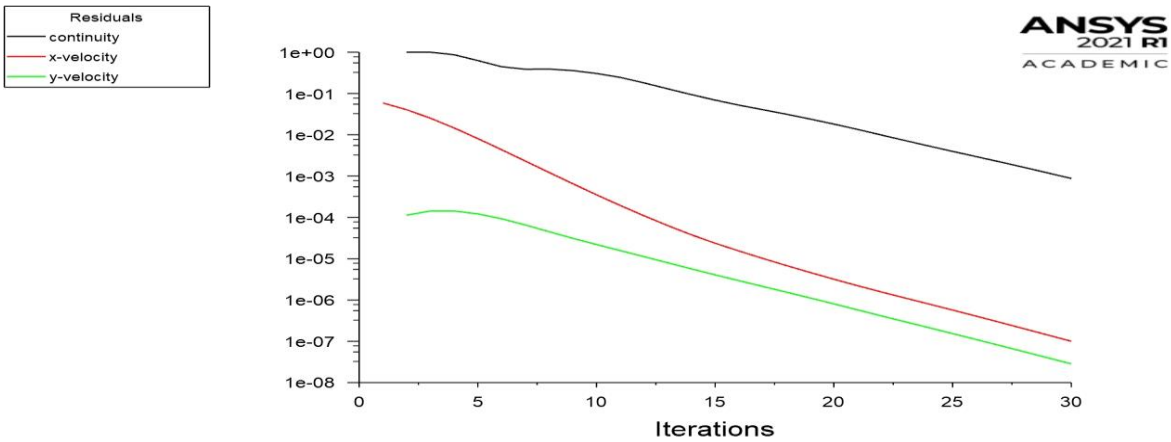


Meshed View- Edge Sizing and Face Sizing
Number of Nodes= 5151 and Number of Elements= 5000



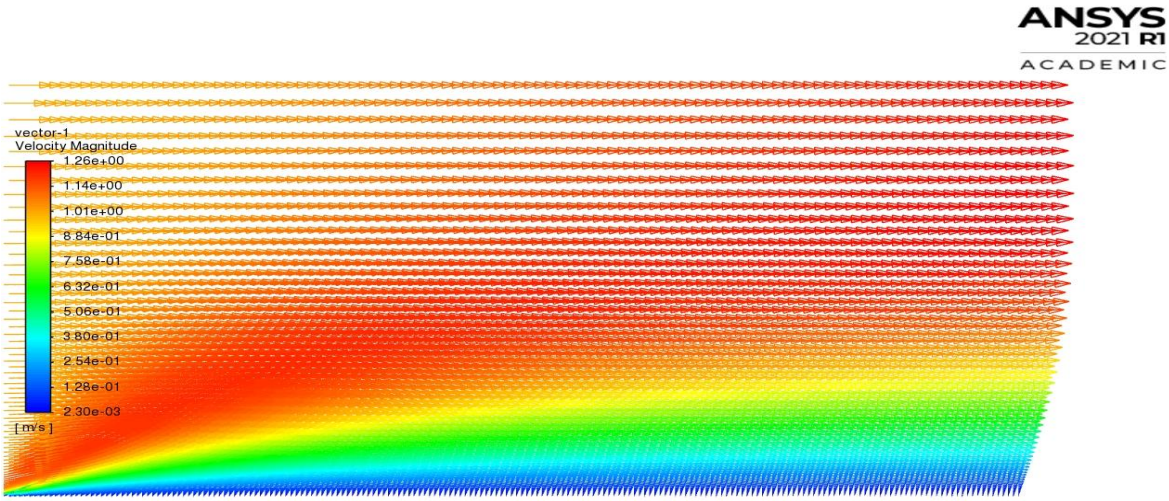
Graphics window of Mesh used for channel flow display inlet (blue) and outlet (red)

❖ Residual



Plot of residuals for continuity, x-velocity and y-velocity

❖ Velocity Vector



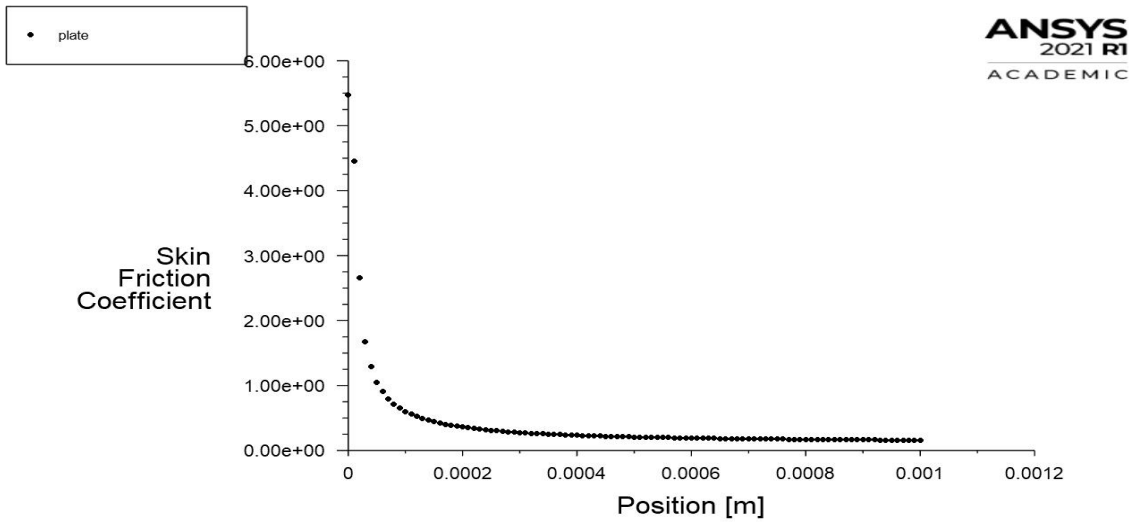
Velocity Graphics from inlet wall to outlet wall.

❖ Forces

```
Forces - Direction Vector (1 0 0)
                          Forces [N]                                        Coefficients
Zone                      Pressure        Viscous         Total             Pressure
Viscous         Total
plate                     0               0.00019268358   0.00019268358     0
0.00038536715   0.00038536715
------------------------- --------------- --------------- --------------- ---------------
--------------- ---------------
Net                       0               0.00019268358   0.00019268358     0
0.00038536715   0.00038536715
```

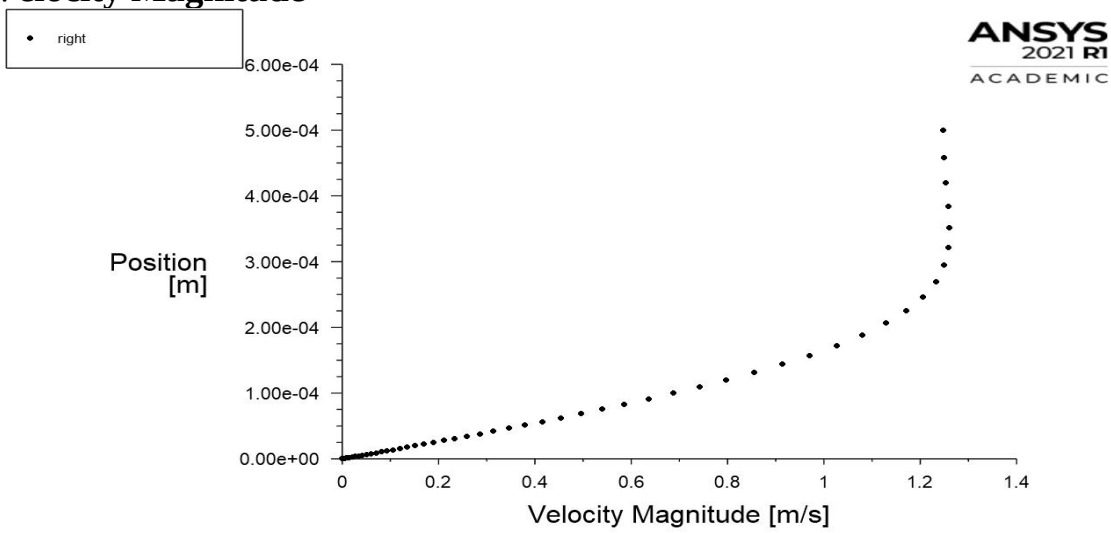
Amount of viscous force along plate for direction vector (1,0,0)

❖ Skin Friction Coefficient



Plot of Skin Friction for V=1 m/s and Density = 1 kg/m^3

❖ Velocity Magnitude



Graph Displays velocity vector from right to the left of the Plane.